

NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) specializing in silicon validation and hardware/software co-design. Built an automated co-simulation system that generates RTL, validates against simulation, and root-causes failures via VCD waveform analysis — scoring **37/92 on CVPD** and **5/5 on ASU Spec2Tapeout**. Designed a GCN ASIC accelerator end-to-end (Synopsys DC + Cadence Innovus, ASAP7 7nm) and extended gem5's C++ cache subsystem to correlate architectural performance models.

TECHNICAL SKILLS

RTL Design & Verification: SystemVerilog | Verilog | Cocotb | SymbiYosys | iverilog | VCD Waveform Analysis | SDC Constraints | Equivalence Checking

Silicon Tools & Flows: Synopsys Design Compiler | Cadence Innovus | Voltus | OpenROAD | Yosys | ASAP7 7nm FinFET | SkyWater 130nm | Static Timing Analysis

Programming & Systems: C++ | Python | SystemVerilog | gem5 Architectural Simulation | MATLAB/Simulink | Docker | Linux | Ansys Maxwell FEA

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Arizona State University

May 2026

Tempe, AZ

Coursework: VLSI Design · VLSI Design Automation · Advanced Computer Architecture · Algorithm/Hardware Co-Design · SoC Design

Bachelor of Engineering — Electrical and Electronics

BMS College of Engineering

August 2022

Bengaluru, India

PROJECT EXPERIENCE

SiliconCrew — Automated HW/SW Co-Simulation & Silicon Validation System

Nov 2025 – Present

- Built SiliconCrew, an automated hardware/software integration system (21 tools, Python + C++) that drives a complete silicon validation pipeline — RTL generation → lint → simulation → Yosys synthesis → OpenROAD place-and-route → post-synthesis simulation → GDSII — across SkyWater 130nm and ASAP7 7nm PDKs; integrates SymbiYosys formal verification and Cocotb constrained-random testbenches.
- Automates root-cause debug across the hardware/software boundary: on simulation failure, reads VCD waveforms, traces signal-level faults back to RTL or configuration errors, patches and re-verifies — replicating the diagnose-fix-revalidate cycle of silicon bringup; enforces SDC constraint validation, RTL-vs-netlist equivalence checking, and signoff guardrails before GDSII.
- Achieved **37/92 first-pass on CVPD** (Comprehensive Verilog Design Problems) and **5/5 on ASU Spec2Tapeout** with zero human intervention.

GCN ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm

Spring 2025

- Designed an 11-module GCN ASIC accelerator in SystemVerilog — 8 parallel MACs for 96-element dot products and a 12-state FSM for bidirectional COO edge aggregation — and completed the full RTL-to-GDSII flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT (1 ns clock, 10,878 cells), Cadence Innovus APR with CTS, and Voltus post-APR power analysis driven by VCD activity files.
- Achieved 50,300 μm^2 chip area at 50.6% core density, 52.29 mW total power (64.8% switching), and 99.2 ns end-to-end latency.

5-Stage Pipelined MIPS Processor — SystemVerilog

Fall 2025 – Spring 2026

- Implemented a 5-stage pipelined MIPS processor in SystemVerilog (33 instructions) with a 12-path forwarding network (EX-to-EX, MEM-to-EX, WB-to-ID, branch, JR) and 6 hazard stall conditions; extended with an 8-entry fully-associative CAM instruction cache (3-state FIFO FSM) and LL/SC atomic operations with link-address tracking.

Static Timing Analysis Engine — Python

Fall 2025 – Spring 2026

- Built a complete Static Timing Analysis engine in Python from scratch — Liberty NLDM parser (7×7 delay/slew LUTs with bilinear interpolation), .bench circuit-graph parser, topological sort via Kahn's algorithm, forward arrival-time and slew propagation, backward RAT/slack computation, and critical path extraction; validated against ISCAS benchmarks c17, b15, and c7552.

LRU-IPV Cache Replacement Policy — gem5 C++

Fall 2024

- Extended gem5's C++ architectural simulator with a new LRU-IPV cache replacement policy as a first-class SimObject (4 cache interface methods across 5 source files); used controlled experiments across 3 L2 configurations (128/256/512 kB) to isolate capacity-vs-policy performance effects, correlating replacement behavior ($637 \rightarrow 112 \rightarrow 2$) with IPC at 41.8M instructions.

WORK EXPERIENCE

Systems Engineer

NCR GOLD

Oct 2022 – Apr 2024

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

Research Intern — Propulsion & Power Electronics

Indian Institute of Science

Oct 2021 – Aug 2022

Bengaluru, India

- Validated motor hardware designs via Ansys Maxwell FEA simulation (1.5 kW axial flux BLDC) — iterated on stator geometry and magnetic materials to improve flux linkage and reduce core losses; modeled FOC firmware control algorithms and battery management in MATLAB/Simulink, correlating simulation results with hardware design changes for eVTOL propulsion reliability.

AWARDS & HONORS

- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.